

Independence $\Rightarrow$ Betweenness

Result — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Premises

- **Independence.** For all $p, q, r \in \Delta(X)$ and $\lambda \in (0,1]$: $p \succcurlyeq q$ if and only if $\lambda p + (1-\lambda)r \succcurlyeq \lambda q + (1-\lambda)r$.

0.2 Conclusion

- **Betweenness.** For all $p, q \in \Delta(X)$ and $\lambda \in (0,1)$: if $p \succ q$ then $p \succ \lambda p + (1-\lambda)q \succ q$, and if $p \sim q$ then $p \sim \lambda p + (1-\lambda)q$.

0.3 Proof

Fix $\lambda \in (0,1)$ and write $m = \lambda p + (1-\lambda)q$. Independence with $r = p$ and mixing weight $1-\lambda$ gives $p \succcurlyeq q \iff (1-\lambda)p + \lambda p \succcurlyeq (1-\lambda)q + \lambda p$, i.e. $p \succcurlyeq q \iff p \succcurlyeq m$, and likewise $q \succcurlyeq p \iff m \succcurlyeq p$. Hence $p \succ q \iff p \succ m$ and $p \sim q \iff p \sim m$. Independence with $r = q$ and weight λ gives $p \succcurlyeq q \iff m \succcurlyeq q$ and $q \succcurlyeq p \iff q \succcurlyeq m$, so $p \succ q \iff m \succ q$. Together: $p \succ q \implies p \succ m \succ q$ and $p \sim q \implies p \sim m$.

0.4 Notes

No ordering assumptions are needed because independence is taken in the biconditional form.

Record: topics/decision-theory/results/independence-implies-betweenness.yaml